

Notes: Treﬂer (AER, 2004)

The Long and Short of the Canada-U.S. Free Trade Agreement

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1 Big picture

- **Question.** What were the short-run adjustment costs and long-run efficiency gains of the Canada-U.S. Free Trade Agreement (FTA) for Canadian manufacturing?
- **Core empirical setting.** The paper studies the Canada-U.S. FTA, implemented on January 1, 1989, as a relatively clean reciprocal trade-liberalization episode between two industrialized economies.
- **Main policy variables.** The paper constructs FTA-mandated tariff concessions: Canadian tariff cuts granted to the United States and U.S. tariff cuts granted to Canada. These are measured at a highly disaggregated Canadian 4-digit SIC industry level.
- **Main outcomes.** The paper studies employment, production and nonproduction labor, labor productivity, earnings, wages, hours, inequality, output, import prices, trade creation, and trade diversion.
- **Headline employment result.** In the most impacted import-competing industries, Canadian tariff concessions reduced employment by about 12 percent. For manufacturing as a whole, the FTA reduced employment by about 5 percent, roughly 100,000 jobs.
- **Headline productivity result.** In the most impacted import-competing industries, industry-level labor productivity rose by about 15 percent. In the most impacted export-oriented industries, plant-level labor productivity rose by about 14 percent.
- **Mechanism.** Import-side productivity gains mainly reflect the exit or contraction of low-productivity plants and market-share reallocation toward more productive producers. Export-side gains appear within plants that benefit from better access to the U.S. market.
- **Welfare interpretation.** The FTA created more trade than it diverted and did not raise import prices; if anything, import prices may have fallen. These facts support the conclusion that the FTA likely improved aggregate welfare.
- **Economic tension.** The paper's title captures the key trade-off: short-run losses were concentrated among workers and low-productivity plants, while long-run gains accrued to consumers, efficient plants, and users of intermediate inputs.

2 Incremental contribution of the paper

- **From one-sided trade evaluation to joint cost-benefit evaluation.** Earlier work often studied either worker displacement or productivity gains. This paper studies both within one unified empirical design.
- **Industrialized-country evidence.** Much prior econometric evidence on trade liberalization and productivity came from developing countries. This paper provides clean evidence from Canada, a high-income industrialized economy.
- **Clean policy experiment.** The FTA was not bundled with a broad macroeconomic stabilization, market reform, or financial crisis. This makes it easier to isolate trade-policy effects.
- **Direct tariff measurement.** The paper uses tariff concessions rather than broad openness proxies, addressing a major criticism of the trade-liberalization literature.

- **Reciprocal liberalization.** The FTA created both import competition and export-market access. This allows the paper to distinguish effects of Canadian tariff cuts from effects of U.S. tariff cuts.
- **Plant-level and industry-level evidence.** By combining industry and plant data, the paper separates within-plant effects from market-share reallocation effects.
- **Methodological improvement over earlier FTA studies.** The long-difference difference-in-differences approach avoids dynamic panel problems caused by annual autocorrelation in employment and output.
- **Welfare bridge.** The paper links reduced-form empirical estimates to welfare-relevant conditions: whether trade creation dominates trade diversion and whether import prices rise.

3 Institutional background and policy variation

3.1 The Canada-U.S. Free Trade Agreement

- The FTA began in 1989 and reduced bilateral tariffs between Canada and the United States.
- The agreement reduced tariffs against the partner country but did not reduce tariffs against the rest of the world. This creates within-industry policy variation between partner and nonpartner tariff paths.
- Before the FTA, Canada had meaningful tariff protection in some sectors. Average Canadian manufacturing tariffs against the United States were about 8.1 percent in 1988, and one in four Canadian industries had tariffs above 10 percent.
- The U.S. also reduced tariffs against Canada, though U.S. initial tariffs were lower than Canadian tariffs.

3.2 Policy variables

Let industry be indexed by i and year by t . Treffler defines two central policy levers:

- τ_{it}^{CA} : FTA-mandated Canadian tariff concessions granted to the United States.
- τ_{it}^{US} : FTA-mandated U.S. tariff concessions granted to Canada.

For the FTA period, the tariff change is

$$\Delta\tau_{i1}^k = \frac{\tau_{i,1996}^k - \tau_{i,1988}^k}{1996 - 1988}, \quad k \in \{\text{CA}, \text{US}\}. \quad (1)$$

The baseline year is 1988 because tariff reductions start on January 1, 1989. Thus 1988 is the pre-treatment baseline.

3.3 Import-competing and export-oriented exposure

- The most impacted import-competing group consists of the one-third of industries with the largest Canadian tariff concessions.
- The most impacted export-oriented group consists of the one-third of industries with the largest U.S. tariff concessions.

- Canadian tariff cuts mainly intensify import competition from U.S. firms.
- U.S. tariff cuts mainly improve Canadian access to the U.S. market.

4 Data

4.1 Industry data

- The industry analysis uses Canadian manufacturing industries at the 4-digit Canadian SIC level.
- The sample contains roughly 213 industries after data harmonization and aggregation decisions.
- Canadian data come from the Annual Survey of Manufactures, Labour Force Survey, Statistics Canada trade data, input-output data, price data, and concordances.
- U.S. data come from the NBER Manufacturing Productivity Database, Feenstra trade data, and updates by the author.

4.2 Plant-level data

- Plant-level analysis uses continuing plants observed in 1980, 1986, 1988, and 1996.
- The main plant sample contains 3,801 plants for employment and slightly fewer for some outcomes.
- Plant controls include log 1980 employment, log 1980 earnings per worker, log 1980 labor productivity, log plant age, and a dummy for whether the plant is older than seven years in 1980.
- The plant data mostly cover larger long-form plants. Appendix E argues that selection into long-form plants is not central for the paper's main conclusions.

4.3 Outcomes

The paper studies a broad set of outcomes:

- employment;
- production-worker and nonproduction-worker employment;
- labor productivity;
- annual earnings, hourly wages, and hours;
- earnings inequality between nonproduction and production workers;
- gross output per plant;
- imports from the United States and from the rest of the world;
- import prices and import quantities.

5 Econometric design

5.1 Period-level outcome changes

Let Y_{it} be an industry-level outcome. Define the average annual log change during the FTA period as

$$\Delta y_{i1} = \frac{\ln Y_{i,1996} - \ln Y_{i,1988}}{1996 - 1988}, \quad (2)$$

and the pre-FTA average annual log change as

$$\Delta y_{i0} = \frac{\ln Y_{i,1986} - \ln Y_{i,1980}}{1986 - 1980}. \quad (3)$$

The FTA-period comparison uses 1988 because 1988 is the last year before the first tariff cut.

5.2 Basic industry specification

The conceptual regression is

$$\Delta y_{is} = \theta_s + \beta^{\text{CA}} \Delta \tau_{is}^{\text{CA}} + \beta^{\text{US}} \Delta \tau_{is}^{\text{US}} + \varepsilon_{is}, \quad s = 0, 1. \quad (4)$$

This equation asks whether industries with deeper tariff concessions experienced different changes in outcomes.

5.3 Secular industry growth control

Because politically protected industries may have different pre-existing trends, the paper adds an industry fixed effect α_i :

$$\Delta y_{is} = \alpha_i + \theta_s + \beta^{\text{CA}} \Delta \tau_{is}^{\text{CA}} + \beta^{\text{US}} \Delta \tau_{is}^{\text{US}} + \varepsilon_{is}. \quad (5)$$

Differencing across periods removes α_i and controls for long-run industry-specific secular growth.

5.4 Industry-specific shocks and business conditions

The baseline specification also controls for:

- $\Delta y_{is}^{\text{US}}$, the corresponding U.S. industry outcome, used to absorb industry-specific demand and supply shocks common to Canada and its major trading partners;
- Δb_{is} , an industry-specific business-conditions control generated from manufacturing GDP and the real exchange rate.

The baseline difference-in-differences equation is

$$\Delta y_{i1} - \Delta y_{i0} = \theta + \beta^{\text{CA}} (\Delta \tau_{i1}^{\text{CA}} - \Delta \tau_{i0}^{\text{CA}}) + \beta^{\text{US}} (\Delta \tau_{i1}^{\text{US}} - \Delta \tau_{i0}^{\text{US}}) \quad (6)$$

$$+ \gamma (\Delta y_{i1}^{\text{US}} - \Delta y_{i0}^{\text{US}}) + \delta (\Delta b_{i1} - \Delta b_{i0}) + v_i. \quad (7)$$

5.5 Plant-level specification

For plant k in industry i , the plant-level counterpart is

$$\Delta y_{ik1} - \Delta y_{ik0} = \theta + \beta^{\text{CA}} (\Delta \tau_{i1}^{\text{CA}} - \Delta \tau_{i0}^{\text{CA}}) + \beta^{\text{US}} (\Delta \tau_{i1}^{\text{US}} - \Delta \tau_{i0}^{\text{US}}) \quad (8)$$

$$+ \gamma (\Delta y_{i1}^{\text{US}} - \Delta y_{i0}^{\text{US}}) + \delta (\Delta b_{i1} - \Delta b_{i0}) + \phi' X_{ik,1980} + v_{ik}. \quad (9)$$

This specification identifies how continuing plants respond to tariff concessions, conditional on initial plant characteristics.

5.6 Why long differences matter

The paper emphasizes that prior annual-panel FTA studies may suffer from autocorrelation problems. Employment and output display strong autocorrelation at annual frequencies. By using long differences between pre-FTA and FTA periods, Trefler avoids dynamic panel estimation problems and obtains more reliable standard errors.

5.7 Endogeneity and IV

The paper considers instrumental variables using 1980 industry characteristics such as wages, employment, imports from the United States, and U.S. imports from Canada, including squares and cross-products. The IV estimates are reported, but the author downplays them because Hausman tests generally reject endogeneity. The main interpretation relies on the OLS long-difference estimates with extensive controls.

6 Main employment results

6.1 Aggregate employment effects

- In the baseline industry-level specification, Canadian tariff concessions reduced employment by 12 percent in the most impacted import-competing industries.
- The total FTA impact on all Canadian manufacturing employment is about -5 percent.
- The total employment loss corresponds to roughly 100,000 jobs.
- The losses are concentrated in the heavily protected import-competing industries.

6.2 Import competition versus export access

- Canadian tariff concessions caused substantial employment contraction in import-competing industries.
- U.S. tariff concessions had weaker and less robust employment effects.
- At the plant level, the U.S. tariff concessions had little effect on employment on average, likely because positive exporter responses and negative nonexporter responses offset each other.

6.3 Economic interpretation

The employment results show the short-run adjustment cost of free trade. The FTA forced resources out of low-end, previously protected industries. The paper argues that these costs were probably short-run in the sense that total Canadian employment recovered, but the workers and plants directly affected by the transition bore large concentrated losses.

7 Main productivity results

7.1 Industry-level productivity gains from import competition

- The baseline industry estimate implies that Canadian tariff concessions raised labor productivity by about 15 percent in the most impacted import-competing industries.

- This corresponds to a compound annual growth rate of roughly 1.9 percent.
- The plant-level effect is smaller and only marginally significant, implying that much of the industry-level productivity gain came from reallocation: low-productivity plants contracted or exited, while high-productivity plants gained market share.

7.2 Plant-level productivity gains from export access

- U.S. tariff concessions raised plant-level labor productivity by about 14 percent in the most impacted export-oriented industries.
- The corresponding industry-level effect is smaller because new or young exporters may enter with lower productivity, reducing average industry productivity.
- The plant-level productivity gain indicates that access to the U.S. market improved performance among continuing Canadian plants.

7.3 Total manufacturing productivity effect

The FTA raised labor productivity in manufacturing as a whole by roughly 6-7 percent. This is large because many industries were already duty-free before the FTA, so the average effect is driven by large responses in affected industries.

7.4 Labor productivity versus TFP

Because the Canadian ASM lacks plant-level capital stock and investment data, the paper cannot directly estimate plant-level TFP. Trefler argues that the labor-productivity gains likely reflect substantial TFP gains because there is little evidence that the results are driven by capital deepening, more intensive intermediate input use, or rising markups. A rough interpretation is that FTA-induced TFP gains are about half of the labor-productivity gains.

8 Trade creation, trade diversion, prices, and welfare

8.1 Trade creation condition

For preferential trade agreements, welfare gains are not automatic. Trefler uses a Krishna-style condition comparing trade creation and trade diversion. Let $\Delta \ln m_{i1}^{\text{US}}$ be the change in Canadian imports from the United States and $\Delta \ln m_{i1}^{\text{ROW}}$ be the change in imports from the rest of the world. A sufficient condition is approximately

$$-0.8 \frac{\Delta \ln m_{i1}^{\text{US}}}{\Delta \tau_{i1}^{\text{US}}} - 0.2 \frac{\Delta \ln m_{i1}^{\text{ROW}}}{\Delta \tau_{i1}^{\text{US}}} > 0, \quad (10)$$

where 0.8 is the U.S. share of Canadian imports.

8.2 Empirical result

- Canadian tariff concessions raised Canadian imports from the United States by about 54 log points.
- Canadian tariff concessions lowered Canadian imports from the rest of the world by about 40 log points.

- Plugging the estimates into the welfare condition gives a positive and statistically significant number.
- Therefore, FTA trade creation dominated trade diversion sufficiently to satisfy the welfare condition.

8.3 Import prices

- A preferential trade agreement is less likely to be welfare-improving if it raises import prices.
- The paper examines import unit values at the 10-digit Harmonized System level.
- The OLS estimate suggests no increase in import prices.
- The IV estimate suggests that the FTA may have reduced import prices by about 7 percent among the most affected products.

8.4 Welfare implication

Because trade creation dominated trade diversion and import prices did not rise, the paper concludes that the FTA likely increased aggregate Canadian welfare, even though it imposed large adjustment costs on particular workers and plants.

9 Labor-market composition, earnings, and inequality

9.1 Production and nonproduction workers

- Canadian tariff concessions reduced production-worker employment strongly.
- Nonproduction-worker employment was less affected at the industry level but fell at the plant level in some specifications.
- The FTA induced skill upgrading at the industry level: the ratio of nonproduction to production workers increased.
- The skill-upgrading result appears more at the industry level than at the plant level, suggesting market-share shifts toward nonproduction-worker-intensive plants.

9.2 Earnings and wages

- Contrary to popular fears, the FTA did not reduce Canadian earnings or wages.
- Annual earnings rose modestly for all workers, especially at the plant level.
- The increase came mainly from hourly wage increases rather than hours worked.
- The wage effects are statistically significant but economically small.

9.3 Why did wages not fall?

The paper discusses several possible explanations:

- **Union bargaining:** not strongly supported because unionization is not meaningfully correlated with tariff concessions.

- **Skill upgrading:** possible, but evidence is tentative and sensitive to the clothing industry.
- **Seniority-based attrition:** best supported by the data. Industries with deeper tariff cuts experienced larger increases in current job tenure, suggesting that less senior workers exited disproportionately.

9.4 Inequality

There is no evidence that the FTA increased earnings inequality between nonproduction and production workers. The estimated effects on the nonproduction-to-production earnings ratio are close to zero.

10 What explains rising labor productivity?

10.1 Scale economies

The paper tests whether plants moved down average cost curves by estimating effects on output per plant. The evidence is weak: import-competing plants contracted and export-oriented plants expanded, but the estimates are not strongly significant. Therefore, simple scale effects are not the main explanation.

10.2 Plant rationalization

A more plausible explanation is rationalization: Canadian plants may have produced fewer product lines and received more specialized mandates within North American production networks. This is consistent with related evidence that foreign-owned Canadian plants reduced product diversification as exports rose.

10.3 Selection and reallocation

For import-competing industries, the main mechanism is selection: low-productivity plants contracted or disappeared, raising industry-level productivity. This is the central allocative-efficiency channel.

10.4 Within-plant productivity improvement

For export-oriented plants, the evidence points more to within-plant productivity improvement because the plant-level productivity effects of U.S. tariff concessions are large.

11 Interpretation and limitations

11.1 What the paper identifies well

- It provides strong reduced-form evidence that tariff concessions caused large changes in employment and productivity.
- It distinguishes import-competition and export-access channels.
- It separates industry-level reallocation effects from continuing-plant effects.
- It connects empirical results to welfare-relevant trade creation, diversion, and price conditions.

11.2 Key limitations

- Tariff cuts were not randomly assigned, so the identification relies on controls, long differencing, and robustness checks rather than experimental variation.
- The plant-level data cover continuing long-form plants and therefore do not directly capture all entry and exit.
- The main productivity measure is labor productivity, not plant-level TFP.
- The paper is intentionally reduced-form, so it does not fully model general-equilibrium feedbacks or welfare distribution.

11.3 Why the limitations do not overturn the main message

The results are robust across many specifications, including alternative business-cycle controls, alternative U.S. controls, deletion of outliers, deletion of the auto sector, plant-level controls, and IV estimates. The broad pattern is stable: employment losses are concentrated in protected import-competing sectors, while productivity gains are large and economically meaningful.

12 Tables and figures

12.1 Figure 1: Canadian and U.S. bilateral tariffs in manufacturing

Read:

- The figure plots Canadian tariffs against the United States and rest of world, and U.S. tariffs against Canada and rest of world.
- The FTA created a clear wedge between partner-country tariffs and rest-of-world tariffs.
- The Canadian tariff cut was larger than the U.S. tariff cut because Canadian tariffs were initially higher.

Economic message: The figure visually identifies the FTA policy experiment: bilateral liberalization without equivalent multilateral liberalization.

12.2 Figure 2 and Table 1: Business conditions and employment effects

Read Figure 2:

- The figure plots real Canadian manufacturing GDP and shows the early-1990s recession during the FTA period.
- Treffer chooses the pre-FTA period partly so that business-cycle patterns are comparable across pre-FTA and FTA periods.

Read Table 1:

- Baseline industry OLS estimates imply a 12 percent employment loss in the most impacted import-competing industries.
- The total FTA impact is around -5 percent for manufacturing employment.
- Plant-level estimates are very similar for Canadian tariff concessions, suggesting that industry results partly reflect within-continuing-plant contraction.

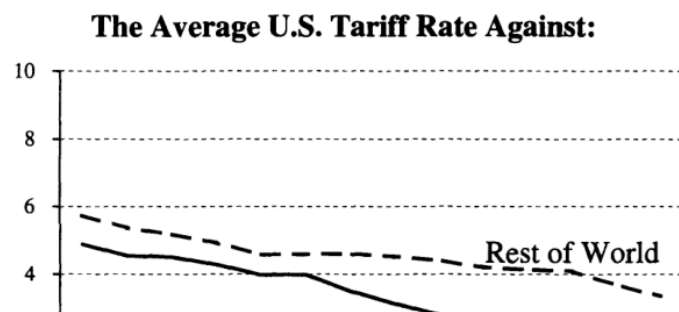
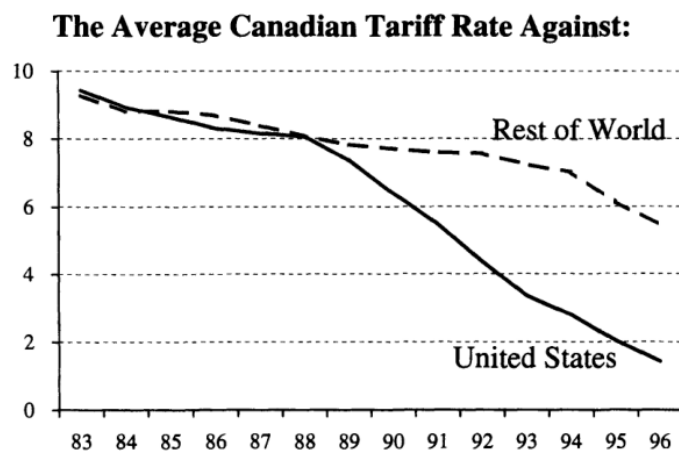
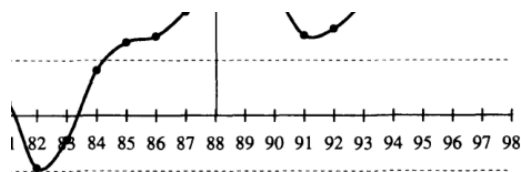


Figure 1: Canadian and U.S. bilateral tariffs in manufacturing.

- Robustness rows show that the result is not driven by the business-cycle control, U.S. control, outliers, or auto industries.

Economic message: The FTA imposed large, concentrated employment losses in protected import-competing industries.



2. REAL CANADIAN MANUFACTURING GDP

t fa4ctor cost, 1992 dollars.

and shortly. Introducing Δb_{is} and Δy_{is}^{US} in equation (4) yields

$$\ln y_{it} = \alpha + \beta \Delta b_{is} + \gamma \Delta y_{is}^{US} + \epsilon_{it}$$

(a) Figure 2: Real Canadian manufacturing GDP.

TABLE 4. <i>U.S. IMPORTS: ESTIMATES FROM A BIVARIATE SYSTEM</i>												
	Construction of Δb	Canadian tariffs Δy^{CA}		U.S. tariffs Δy^{US}		Business conditions Δb		U.S. control Δy^{US}		Adjusted R^2	Overall/Hausman	Total FTA impact
		β^{CA}	t	β^{US}	t	δ	t	γ	t			TFI
Industry level, OLS												
1	gdp, rer (2)	-0.12	-2.35	-0.03	-0.67	0.29	6.96	0.15	2.21	0.24		-0.05 -2.66
2	gdp, rer (0)	-0.11	-2.03	-0.04	-0.91	0.30	3.66	0.21	2.75	0.12		-0.06 -2.58
3	gdp (2)	-0.11	-2.08	-0.03	-0.66	0.37	6.60	0.15	2.16	0.23		-0.05 -2.41
4	—	-0.14	-2.40	-0.02	-0.52			0.20	2.58	0.07		-0.06 -2.58
5	gdp, rer (2)	-0.13	-2.48	-0.02	-0.39	0.28	6.74	0.29	3.00	0.24		-0.05 -1.71
6	gdp, rer (2)	-0.14	-2.75	-0.03	-0.80	0.30	7.12			0.23		-0.06 -3.16
7	—	-0.17	-2.88	-0.03	-0.66					0.04		-0.07 -3.15
8	gdp, rer (2)	-0.14	-2.24	-0.02	-0.53	0.29	6.89	0.15	2.11	0.24		-0.06 -2.65
9	gdp, rer (2)	-0.12	-2.30	-0.06	-1.45	0.30	7.23	0.14	2.04	0.27		-0.06 -3.24
Plant level, OLS												
10	gdp, rer (2)	-0.12	-3.76	0.00	0.15	0.13	4.59	0.25	5.29	0.04		-0.04 -3.26
11	gdp, rer (2)	-0.12	-3.60	-0.01	-0.26	0.16	5.63	0.25	5.21	0.02		-0.04 -3.51
Industry level, IV												
12	gdp, rer (2)	-0.24	-1.45	0.09	0.66	0.29	6.68	0.15	2.06	0.22	0.60/0.65	-0.04 -1.26
13	gdp, rer (2)	-0.24	-1.43	0.04	0.29	0.31	6.37	-0.16	-0.50	0.20	0.67/0.57	-0.05 -1.57
Plant level, IV												
14	gdp, rer (2)	-0.19	-2.40	0.07	0.94	0.13	4.30	0.24	4.96	0.04	0.140/0.99	-0.04 -2.55
15	gdp, rer (2)	-0.19	-2.44	0.07	0.92	0.13	4.17	0.16	0.95	0.03	0.100/0.89	-0.04 -3.10

Notes: The dependent variable is the log of employment. The estimating equation is equation (6) for the industry-level regressions and equation (7) for the plant-level regressions. β^{CA} is scaled so that it gives the log-point impact of the Canadian tariff concessions on employment in the most impacted, import-competing industries. β^{US} is scaled so that it gives the log-point impact of the U.S. tariff concessions on employment in the most impacted, export-oriented industries. The "Total FTA impact" column gives the joint impact of the tariff concessions on employment in all 213 industries. The "Overall/Hausman" column reports p -values for the overidentification and Hausman tests. Rejection of the instrument set or exogeneity are indicated by p -values less than 0.01. The number of observations is 213 for the industry-level regressions and 3,801 for the plant-level regressions. In rows 4 and 7, the business conditions variable is omitted so that business conditions are controlled for implicitly by double-differencing $\Delta y_{it} - \Delta y_{it}^{US}$. In row 5 the U.S. control is replaced by the Japan-U.K. control discussed in the text. In row 8, the 2 "outlier" observations with the largest Canadian tariff cuts are omitted. In row 9, all 9 observations associated with the automotive sector are omitted. In row 11, the plant controls are omitted. In rows 12 and 14, only the Canadian and U.S. tariff variables are instrumented. In rows 13 and 15, the two tariff variables and the U.S. control are instrumented.

(b) Table 1: Detailed employment results.

Figure 2 and Table 1: Business conditions and employment effects.

12.3 Table 2 and Table 3: Labor productivity, trade creation, diversion, and prices

Read Table 2:

- Canadian tariff concessions raise industry-level labor productivity by 15 percent in the most impacted import-competing industries.
- U.S. tariff concessions raise plant-level labor productivity by 14 percent in the most impacted export-oriented industries.
- Total manufacturing productivity rises by about 6-7 percent.

Read Table 3:

- Canadian imports from the United States rise sharply, while imports from the rest of the world fall.
- This satisfies the trade-creation-versus-trade-diversion welfare condition.
- Import prices do not rise; IV estimates suggest they may fall.
- Import quantities respond strongly, indicating that the product-level price/quantity data are informative rather than pure noise.

Economic message: The FTA generated large productivity gains and met key empirical conditions for welfare improvement.

Construction of Δb	Canadian tariffs $\Delta\tau^{CA}$		U.S. tariffs $\Delta\tau^{US}$		Business conditions Δb		U.S. control $\Delta\gamma^{US}$		Adjusted R^2	Overid/Hausman	Total FTA impact τ
	β^{CA}	t	β^{US}	t	δ	t	γ	t			
Industry level, OLS											
1 <i>gdp, rer</i> (2)	0.15	3.11	0.04	1.14	0.25	8.30	0.16	1.99	0.31		0.058 3.79
2 <i>gdp, rer</i> (0)	0.15	2.77	0.02	0.40	0.13	1.79	0.28	3.05	0.09		0.050 2.87
3 <i>gdp</i> (2)	0.17	3.21	0.04	1.17	0.25	5.19	0.21	2.43	0.18		0.065 3.87
4 —	0.16	2.85	0.01	0.34			0.29	3.23	0.08		0.051 2.89
5 <i>gdp, rer</i> (2)	0.14	2.79	0.05	1.36	0.26	8.77	0.05	0.31	0.29		0.058 2.46
6 <i>gdp, rer</i> (2)	0.14	2.96	0.05	1.44	0.27	8.82			0.30		0.059 3.89
7 —	0.15	2.58	0.03	0.76					0.04		0.053 2.98
8 <i>gdp, rer</i> (2)	0.17	2.97	0.04	0.98	0.26	8.34	0.16	1.95	0.30		0.061 3.76
9 <i>gdp, rer</i> (2)	0.16	3.27	0.02	0.49	0.26	8.61	0.18	2.24	0.33		0.051 3.36
Plant level, OLS											
10 <i>gdp, rer</i> (2)	0.08	1.70	0.14	3.97	0.12	3.95	0.11	1.51	0.06		0.074 4.92
11 <i>gdp, rer</i> (2)	0.09	1.92	0.11	3.02	0.10	3.18	0.14	1.79	0.01		0.066 4.39
Industry level, IV											
12 <i>gdp, rer</i> (2)	0.15	1.10	0.10	0.86	0.26	8.09	0.14	1.53	0.30	0.86/0.43	0.081 3.41
13 <i>gdp, rer</i> (2)	0.13	0.89	0.13	1.01	0.28	6.99	-0.08	-0.28	0.28	0.87/0.51	0.083 3.40
Plant level, IV											
14 <i>gdp, rer</i> (2)	0.22	1.67	0.05	0.49	0.11	3.20	0.17	1.80	0.06	0.06/0.77	0.082 2.53
15 <i>gdp, rer</i> (2)	0.79	2.58	-0.49	-1.73	-0.19	-1.29	2.07	2.29	0.05	0.76/0.52	0.050 0.39

Notes: The dependent variable is the log of labor productivity. The estimating equation is equation (6) for the industry-level regressions and equation (7) for the plant-level regressions. The number of observations is 211 for the industry-level regressions and 3,726 for the plant-level regressions. See the notes to Table 1 for additional details. In rows 4 and 7, the business conditions variable is omitted so that business conditions are controlled for implicitly by double-differencing $\Delta y_{it} - \Delta y_{it}$. In row 5 the U.S. control is replaced by the Japan-U.K. control discussed in the text. In row 8, the two “outlier” observations with the largest Canadian tariff cuts are omitted. In row 9, all nine observations associated with the automotive sector are omitted. In row 11, the plant controls are omitted. In rows 12 and 14, only the Canadian and U.S. tariff variables are instrumented. In rows 13 and 15, the two tariff variables and the U.S. control are instrumented.

thus little alternative but to work with labor productivity. I define labor productivity as value added in production activities per hour worked by production workers.⁹ I deflate using 2-digit

same format as the Table 1 employment results so that I can review it quickly. As in Table 1, endogeneity is always rejected¹¹ and all the industry-level OLS results are statistically significant

Variable	Canadian tariffs		U.S. tariffs		Total FTA impact		Business conditions		Adjusted R^2	Overid/ Hausman	Observations
	β^{CA}	t	β^{US}	t	τ	t	δ	t			
Canadian imports from the United States											
OLS Industry	0.54	-4.67	0.16	-2.16	0.01	0.83	0.22*	0.24			211
IV Industry	2.32	0.80	-0.86	-0.40	-0.15	-0.48	0.30	0.15		NA/0.28	211
Canadian imports from the rest of the world											
OLS Industry	-0.40	2.67	0.08	-0.17	0.03	0.12	0.11*	0.05			211
IV Industry	-1.60	-0.54	1.24	0.48	0.22	0.47	0.08	0.04		NA/0.75	211
Canadian import prices											
OLS Product	-0.004	0.20						0.00			4,700
IV Product	-0.073	2.26						0.00		0.51/0.03	4,700
Canadian import quantities											
OLS Product	0.70	15.12						0.05			4,700
IV Product	1.02	12.68						0.04		0.87/0.00	4,700

Notes: The dependent variable is indicated in bold font at the start of each block of results, e.g., “Canadian imports from the United States.” All dependent variables are in log changes. The estimating equation is equation (6) for the industry-level Canadian imports regressions and equation (10) for the product-level import price and quantity regressions. The business conditions variable is the same as in the Table 1, row 1 baseline specification. The U.S. control is not included because it makes no sense in a bilateral import context. β^{CA} and β^{US} are scaled as described in the notes to Table 1. An asterisk indicates statistical significance at the 1-percent level. The “Overid/Hausman” column reports p -values for the overidentification and Hausman tests. Rejection of the instrument set or exogeneity are indicated by p -values of less than 0.01. Blank entries indicate OLS estimation. The product-level import results use wages, employment, squares, and cross-products as instruments. Based on the overidentification test, the industry-level import results drop the squares and cross-products from the instrument set. It is thus just identified (NA).

(a) Table 2: Detailed labor-productivity results.

(b) Table 3: Trade diversion/creation and import prices.

Table 2 and Table 3: Productivity and welfare-related trade outcomes.

12.4 Table 4 and Table 5: Employment composition, wages, inequality, and output

Read Table 4:

- Production-worker employment falls substantially after Canadian tariff concessions.
- Nonproduction-worker employment is less affected at the industry level.
- The FTA induces skill upgrading at the industry level.

Read Table 5:

- Earnings and hourly wages do not fall; they rise modestly.
- Annual hours do not explain the earnings gains.
- Earnings inequality does not rise.
- Gross output per plant offers limited evidence for scale effects as the main productivity mechanism.

Economic message: The FTA reduced employment but did not depress wages among remaining workers; adjustment occurred mainly through employment and composition rather than wage collapse.

12.5 Figure 3 and Table A3: Tenure adjustment and productivity-measure robustness

Read Figure 3:

- Industries with larger Canadian tariff concessions experienced larger increases in current job tenure.
- This supports seniority-based attrition: less senior workers were more likely to leave in affected industries.

Read Table A3:

Variable	Canadian tariffs		U.S. tariffs		Total FTA impact		Business conditions		U.S. control		Adjusted R^2	Overall Hausman
	β^{CA}	t	β^{US}	t	TFI	t	δ	γ				
Employment—Production workers												
1 Industry	-0.14	-2.44	-0.07	-1.56	-0.08	-3.44	0.37*	0.16	0.33			
4 Industry	-0.13	-1.99	-0.07	-1.36	-0.08	-2.89		0.21	0.07			
6 Industry	-0.16	-2.93	-0.08	-1.71	-0.09	-4.08	0.37*		0.32			
12 Industry	-0.20	-1.28	0.03	0.17	-0.06	-1.60	0.37*	0.16	0.32		0.59/0.70	
10 Plant	-0.09	-2.58	-0.03	-0.87	-0.04	-3.01	0.17*	0.29*	0.04			
Employment—Nonproduction workers												
1 Industry	-0.06	-0.71	0.05	0.79	0.00	0.02	0.36*	0.07	0.26			
4 Industry	-0.07	-0.77	0.05	0.73	-0.00	-0.09		0.14	0.00			
6 Industry	-0.06	-0.79	0.04	0.71	-0.00	-0.12	0.36*		0.26			
12 Industry	0.01	0.06	0.11	0.52	0.05	1.22	0.36*	0.11	0.25		0.18/0.36	
10 Plant	-0.14	-3.02	0.04	1.19	-0.03	-1.72	0.02	0.15	0.01			
Skill upgrading												
1 Industry	0.11	1.41	0.10	1.67	0.08	2.72	0.47*	0.24	0.48			
4 Industry	0.08	0.79	0.11	1.26	0.07	1.81		0.24	0.01			
6 Industry	0.12	1.63	0.10	1.56	0.08	2.82	0.47*		0.48			
12 Industry	0.11	0.50	0.15	0.74	0.10	2.21	0.47*	0.25	0.48		0.11/0.83	
10 Plant	-0.01	-0.30	0.04	1.48	0.01	0.96	0.05*	0.17	0.01			

Notes: The dependent variable is indicated in bold font at the start of each block of results, e.g., "Employment—Production workers." The estimating equation is equation (6) for the industry-level regressions and equation (7) for the plant-level regressions. Row numbers correspond to those in Table 1 so that the reader can refer to Table 1 for details of the specification. Rows 1 and 10 are my baseline specifications. See notes to Table 1 for further details, including the scaling of the β^{CA} and β^{US} . An asterisk indicates statistical significance at the 1-percent level. Skill upgrading is the log of the ratio of nonproduction workers to production workers. All dependent variables are in logs. The number of observations in the industry-level

(a) Table 4: Employment and skill upgrading.

TABLE 1. EARNINGS, WAGES, HOURS, INEQUALITY, AND OUTPUT											
Variable	Canadian tariffs		U.S. tariffs		Total FTA impact		Business conditions		U.S. control		Adjusted R ²
	β^{CA}	t	β^{US}	t	TFI	t	δ	γ			
Earnings—All workers											
1 Industry	0.05	2.43	0.03	1.92	0.03	3.80	0.34*		0.25*	0.20	
10 Plant	0.04	2.92	0.04	3.60	0.03	5.64	0.17*		0.19*	0.03	
Earnings—Production workers											
1 Industry	0.04	2.12	0.00	-0.02	0.02	3.61	0.16*		0.11	0.07	
10 Plant	0.05	3.25	0.03	2.57	0.03	4.74	0.12		0.21	0.02	
Earnings—Nonproduction workers											
1 Industry	0.01	0.30	-0.01	-0.29	0.00	0.02	0.18*		0.12	0.08	
10 Plant	0.04	1.48	0.06	2.87	0.03	3.67	0.11		0.11	0.01	
Hourly wages of production workers											
1 Industry	0.05	3.15	0.03	1.84	0.03	4.37	0.60*		0.13	0.33	
10 Plant	0.06	3.23	0.02	1.40	0.03	4.04	0.20		0.16*	0.01	
Annual hours of production workers											
1 Industry	-0.01	-0.48	-0.02	-1.75	-0.01	-1.94	0.02		0.14	0.01	
10 Plant	-0.02	-0.90	0.01	0.80	0.00	-0.12	0.03		0.07	0.00	
Earnings inequality											
1 Industry	-0.04	-1.32	-0.01	-0.55	-0.02	-1.66	0.42*		0.05	0.21	
10 Plant	-0.01	-0.46	0.02	0.97	0.00	0.41	0.13*		0.08	0.00	
Gross output per plant in production activities											
1 Industry	-0.05	-0.65	0.03	0.54	0.00	-0.05	0.30*		0.18		
10 Plant	-0.05	-1.36	0.06	2.01	0.01	0.72	0.16*		0.05		

Notes: The dependent variable is indicated in bold font at the start of each block of results, e.g., "Earnings—All workers." The estimating equation is equation (6) for the industry-level regressions and equation (7) for the plant-level regressions. Row numbers correspond to those in Table 1 so that the reader can refer to Table 1 for details of the specification. Rows 1 and 10 are my baseline specifications. See notes to Table 1 for further details, including the scaling of the β^{CA} and β^{US} . An asterisk indicates statistical significance at the 1-percent level. Earnings inequality is the ratio of nonproduction-worker earnings to production-workers earnings. The U.S. control is not included in the output equations because the published data on the

(b) Table 5: Earnings, wages, hours, inequality, and output.

Table 4 and Table 5: Labor-market composition and earnings.

- Productivity results are robust to alternative definitions of labor productivity.
- Hours adjustment and output deflators are important, but the positive productivity pattern is not an artifact of a single measurement choice.

Economic message: Adjustment occurred through worker composition and plant/industry reallocation, while productivity gains remain robust across measurement choices.

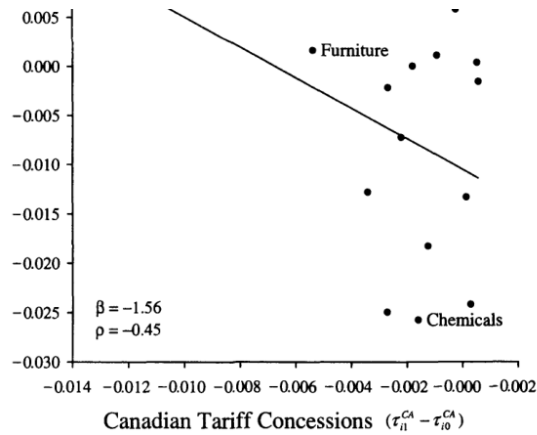


FIGURE 3. CURRENT JOB TENURE CHANGES [(1996–1988)]

(a) Figure 3: Current job tenure changes vs. Canadian tariff concessions.

TABLE A3—SENSITIVITY TO DIFFERENT DEFINITIONS OF LABOR PRODUCTIVITY											
	Canadian tariffs		U.S. tariffs		Total FTA impact		Business conditions		U.S. control		Adjusted R ²
	β^{CA}	t	β^{US}	t	TFI	t	δ	γ			
1. Labor productivity—Production activities—Hours adjusted—Output deflators											
1 Industry	0.15	3.11	0.04	1.14	0.06	3.79	0.25*	0.16	0.31		
10 Plant	0.08	1.70	0.14	3.97	0.07	4.92	0.12*	0.00	0.06		
2. Labor productivity—Production activities—Hours adjusted—Value-added deflators											
1 Industry	0.17	2.96	0.03	0.67	0.06	3.26	0.19*	0.13	0.16		
10 Plant	0.10	2.06	0.16	4.58	0.09	5.69	0.07	0.20*	0.07		
3. Labor productivity—All activities—Not hours adjusted—Output deflators											
1 Industry	0.11	2.27	-0.03	-0.93	0.02	1.29	0.20*	0.24*	0.19		
10 Plant	0.09	2.19	0.13	4.07	0.07	5.54	0.11*	0.13	0.09		

Notes: The dependent variable is indicated in bold font at the start of each block of results. The estimating equation is equation (6) for the industry-level regressions and equation (7) for the plant-level regressions. Rows 1 and 10 are my baseline specifications as in Table 1. See the notes to Table 1 for further details, including the scaling of the β^{CA} and β^{US} . All estimates

(b) Table A3: Sensitivity to alternative productivity definitions.

Figure 3 and Table A3: Labor composition and productivity robustness.

12.6 Table A1 and Table A2: Most impacted industries and alternative periods

Read Table A1:

- The table lists the 71 most impacted import-competing industries.
- Many are low-wage, labor-intensive sectors such as apparel, textiles, furniture, footwear, and related products.

- This helps explain why employment losses were concentrated among vulnerable industries.

Read Table A2:

- The employment and productivity results are stable across alternative pre-FTA and FTA period definitions.
- The estimates are somewhat smaller when the FTA baseline begins in 1989 because the first adjustment year is missed.

Economic message: The most exposed industries are exactly the protected low-end sectors where adjustment costs are most politically salient, and the main results are not an artifact of one period choice.

SIC	Industry description	$\Delta \epsilon^{CA}$	$\Delta \epsilon^{US}$
1131	Brewery Products Industry	-0.331	-0.012
3271	Shipbuilding and Repair Industry	-0.241	-0.012
1931	Canvas and Related Products Industry	-0.183	-0.008
2433	Men's and Boys' Pants Industry	-0.170	-0.053*
2443	Women's Dress Industry	-0.162	-0.076*
2491	Sweater Industry	-0.159	-0.125*
2451	Children's Clothing Industry	-0.159	-0.031*
2441	Women's Coat and Jacket Industry	-0.157	-0.049*
1993	Household Products of Textile Materials	-0.156	-0.017
2442	Women's Sportswear Industry	-0.154	-0.053*
2494	Hosiery Industry	-0.152	-0.040*
1911	Natural Fibers and Felt Processing	-0.150	-0.041*
2434	Men's and Boys' Shirts and Underwear	-0.147	-0.072*
2432	Men's and Boys' Suits and Jackets	-0.147	-0.065*
2431	Men's and Boys' Coat Industry	-0.143	-0.079*
2493	Glove Industry	-0.140	-0.020
2496	Foundation Garment Industry	-0.137	-0.029*
1712	Footwear Industry	-0.127	-0.082*
2612	Upholstered Household Furniture Industry	-0.112	-0.001
1998	Tire Cord Fabric and Other Textiles Products	-0.108	-0.047*
2611	Wooden Household Furniture Industry	-0.106	-0.002
2499	Other Clothing and Apparel Industries	-0.103	-0.040*
2581	Coffin and Casket Industry	-0.101	-0.004
2495	Fur Goods Industry	-0.097	-0.053*
2444	Women's Blouse and Shirt Industry	-0.094	-0.104*
2649	Other Office Furniture Industries	-0.090	-0.002
1041	Fluid Milk Industry	-0.089	-0.006
1991	Narrow Fabric Industry	-0.089	-0.002
2619	Other Household Furniture Industries	-0.089	-0.012
3761	Soap and Cleaning Compounds Industry	-0.088	-0.032*
1829	Other Spun Yarn and Woven Cloth	-0.088	-0.081*
3242	Commercial Trailer Industry	-0.087	-0.004
3792	Adhesives Industry	-0.084	-0.025*
1713	Luggage, Purse and Handbag Industry	-0.082	-0.073*
2543	Wooden Door and Window Industry	-0.079	-0.039*
1691	Plastic Bag Industry	-0.079	-0.023
3612	Lubricating Oil and Grease Industry	-0.079	-0.004
2641	Metal Office Furniture Industry	-0.079	-0.001
2811	Business Forms Printing Industry	-0.078	-0.016
1921	Carpet, Mat and Rug Industry	-0.078	-0.021
1083	Sugar and Chocolate Confectionery	-0.077	-0.024
3751	Paint and Varnish Industry	-0.073	-0.036*
2542	Wooden Kitchen Cabinets, Vanities	-0.073	-0.002
1141	Wine Industry	-0.071	-0.030*
3771	Toilet Preparations Industry	-0.070	-0.024
3993	Floor Tile, Linoleum and Coated Fabrics	-0.070	-0.045*
2721	Asphalt Roofing Industry	-0.069	-0.044*
3791	Printing Ink Industry	-0.069	-0.017
2492	Occupational Clothing Industry	-0.066	-0.031*
3542	Structural Concrete Products Industry	-0.066	-0.015
3021	Metal Tanks (Heavy Gauge) Industry	-0.066	-0.011
3029	Other Fabricated Structural Metal Products	-0.065	-0.033*
3931	Sporting Goods Industry	-0.065	-0.010
1821	Wool Yarn and Woven Cloth Industry	-0.061	0.004
2733	Paper Bag Industry	-0.061	-0.042*
3243	Non-Commercial Trailer Industry	-0.060	0.009
1621	Plastic Pipe and Pipe Fittings Industry	-0.058	-0.031*
3311	Small Electrical Appliance Industry	-0.058	-0.024

TABLE A1—Continued.

SIC	Industry description	$\Delta \epsilon^{CA}$	$\Delta \epsilon^{US}$
1051	Cereal Grain Flour Industry	-0.057	-0.008
3032	Prefabricated Portable Metal Buildings	-0.057	0.000
2941	Iron Foundries	-0.057	-0.002
1093	Potato Chips, Pretzels and Popcorn	-0.056	0.017
3991	Broom, Brush and Mop Industry	-0.055	-0.040*
2792	Stationery Paper Products Industry	-0.054	-0.013
1052	Prepared Flour Mixes and Cereals	-0.054	-0.021
2819	Other Commercial Printing Industries	-0.052	-0.003
2799	Other Converted Paper Products	-0.051	-0.013
3031	Metal Door and Window Industry	-0.051	-0.032*
2821	Platemaking Typesetting and Bindery	-0.051	-0.012
1012	Poultry Products Industry	-0.051	-0.017
3594	Non-Metallic Mineral Insulation	-0.049	-0.058*

Notes: This table reports 1988–1996 changes in tariff concessions for those industries in the most impacted, import-competing group. An asterisk indicates that the industry is also in the most impacted, export-oriented group of industries.

TABLE A2—DIFFERENT CHOICES OF PRE-FTA AND FTA PERIODS

Variable	Canadian tariffs $\Delta \epsilon^{CA}$		U.S. tariffs $\Delta \epsilon^{US}$	
	β^{CA}	t	β^{US}	t
Employment, OLS				
1980–1986, 1988–1996	-0.12	-2.35	-0.03	-0.67
1980–1988, 1988–1996	-0.09	-2.03	0.00	0.04
1980–1986, 1988–1994	-0.13	-2.35	0.00	0.02
1981–1988, 1989–1996	-0.10	-2.05	0.01	0.14
Productivity, OLS				
1980–1986, 1988–1996	0.15	3.11	0.04	1.14
1980–1988, 1988–1996	0.15	3.35	0.00	0.04
1980–1986, 1988–1994	0.17	2.74	0.01	0.20
1981–1988, 1989–1996	0.12	2.64	-0.04	-1.03

Notes: This dependent variable is shown in Table A1. The calculation controls for seasonality.

(a) Table A1, part 1: Most impacted import-competing industries.

(b) Table A1 continued and Table A2: Alternative period choices.

Appendix tables: Exposure lists and period robustness.

13 Short summary

- Treffer studies the Canada-U.S. FTA as a clean reciprocal trade-liberalization episode.
- The paper uses highly disaggregated tariff concessions to measure policy exposure.
- The empirical design compares FTA-period changes with pre-FTA changes and controls for secular industry trends, industry-specific shocks, and business-cycle sensitivity.
- Canadian tariff cuts reduced employment in the most affected import-competing industries by about 12 percent.
- The total employment effect for manufacturing was about -5 percent, or roughly 100,000 jobs.
- Canadian tariff cuts raised industry labor productivity by about 15 percent in affected import-competing sectors.
- U.S. tariff cuts raised plant-level labor productivity by about 14 percent in affected export-oriented sectors.
- Productivity gains on the import side mainly came from reallocation away from low-productivity plants.
- Wages did not fall, and earnings inequality did not rise.
- The FTA created more trade than it diverted and did not raise import prices, supporting a positive aggregate welfare interpretation.

14 Conclusion

14.1 Core conclusion

Treffer's central conclusion is that the Canada-U.S. FTA generated both substantial short-run adjustment costs and substantial long-run efficiency gains. The costs were concentrated in import-competing industries where employment fell sharply. The gains appeared through large productivity improvements and welfare-relevant trade creation.

14.2 Economic intuition

The economic intuition is selection and reallocation. Import competition forced inefficient protected plants to contract or exit, raising industry-level productivity. Export access allowed efficient Canadian plants to expand and improve productivity. Consumers and efficient firms benefited, while displaced workers and struggling plants bore concentrated losses.

14.3 Incremental contribution revisited

The paper's contribution is not only that it estimates FTA effects. It improves the empirical evaluation of trade liberalization by combining direct tariff measures, plant-level data, industry-level data, productivity outcomes, employment outcomes, and welfare-relevant price and trade-flow tests in one design.

14.4 Implications

The paper implies that economists should not describe free trade only in terms of aggregate gains. Trade liberalization can be welfare-improving while still creating severe concentrated losses. Policy design therefore needs to take adjustment costs seriously, especially for workers in protected low-productivity sectors.

14.5 Takeaway

The best one-sentence takeaway is: the Canada-U.S. FTA made Canadian manufacturing more productive and likely raised welfare, but it did so partly by eliminating jobs and reallocating activity away from protected low-productivity plants.